PAPER_1325 — Stellar Core Convection Threshold

Author: Daniel T. Murphy **Framework:** UQFF — Star-Magic v5.27+ **Tier:** D — Astrophysics / Cosmology Open (CLOSED) **Date:** June 11, 2026 **Calculator:** `calculate_paradox({"paradox": "stellar_convection"})`

Master Expression

`Schwarzschild criterion threshold = Φ_res = 0.84`

Match: STRUCTURAL

Interpretation

Convective zone established when temperature gradient exceeds $\Phi_{res} \times$ adiabatic. Sun has outer 30% convective via this criterion.

UQFF Locked Primitives

- $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{CH} = 9$
- $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$, $\Lambda = 0.00729735$, $F_{TRZ} = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.
